ADAPTIVE CREDIT RISK SCORING WITH CONCEPT DRIFT DETECTION

Detecting concept drift in credit-risk data and recovering model performance through drift-triggered adaptive retraining, with explainability monitored throughout.

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Project Report

Contents

Contents 2

01 Executive Summary 3

02 Motivation 3

03 Problem Statement 3

Research Question 4

04 Related Work and Gaps Addressed 4

05 Research Contributions 4

06 Dataset 5

07 System Architecture and Methodology 5

Phase 1 — Preprocessing 5

Phase 2 — Baseline Training 5

Phase 3 — Drift Detection 6

Phase 4 — Adaptive Retraining and SHAP Analysis 6

Phase 5 — Backtesting and Reporting 6

08 Drift Detection Results 6

09 Cost of Model Staleness 7

10 Adaptive Retraining Results 8

A note on F1 and decision thresholds 8

11 Retraining Strategy Comparison 9

12 Explainability Stability (SHAP) 9

13 Interactive Dashboard 10

14 Limitations 11

15 Conclusion 11

01 Executive Summary

This report documents an end-to-end system for credit-risk scoring that detects concept drift in real lending data and recovers lost predictive performance through targeted retraining, while keeping the model's reasoning explainable across retrains. The work is built and evaluated on the public Lending Club dataset (2015–2018, 919,696 cleaned loan records), and every number reported below comes directly from the project's saved output files — not from illustrative or rounded estimates.

The central finding: a credit-risk model trained once on 2015 data and never retrained loses **9.8% of its AUC-ROC by 2018** (0.7422 → 0.6695). Four independent drift-detection methods — the Population Stability Index (PSI), a Kolmogorov-Smirnov (KS) test on the label, Jensen-Shannon (JS) distance, and a classifier-based two-sample test — agree that this drift is real and grows year over year. An adaptive model retrained on a sliding window of recent years recovers most of the lost performance, with the gain growing as drift accumulates (+0.0182 AUC in 2016, rising to +0.0531 by 2018). A secondary experiment compares three retraining strategies and finds that **recency matters more than volume of data**: retraining on the single most recent year outperforms both a 3-year sliding window and a window that keeps all history. SHAP-based explainability monitoring shows that while overall feature-importance rankings stay statistically stable, the specific features in the top 10 shift measurably as drift accumulates.

The system is delivered as a modular Python pipeline (five sequential phases, MLflow-tracked) and an interactive Streamlit dashboard that presents every result and can run the drift detectors and both models live against a new upload.

02 Motivation

Most deployed credit-risk models are trained once and left in production indefinitely. Borrower behaviour and macroeconomic conditions, however, change continuously — interest-rate regimes shift, underwriting standards loosen or tighten, and the population of applicants itself changes composition over time. A model that scored well at deployment can quietly degrade without triggering any alarm, because nothing about the model's own confidence signals that its assumptions about the world have gone stale.

This is the specific failure mode this project targets: not a one-off accuracy problem, but a *silent* and *compounding* one. The practical question a lender needs answered is not just "how accurate is my model today" but "how would I know if it stopped being accurate, and what would I do about it while still being able to explain a rejected loan application to a regulator or an applicant."

03 Problem Statement

How do we build a credit-scoring system that can (a) detect when the data it is scoring has drifted from the data it was trained on, (b) retrain itself when that happens, using an evaluation that cannot be gamed by comparing on mismatched test sets, and (c) remain explainable after retraining, so that a human reviewer can still see why a decision was made and trust that the reasoning has not silently changed.

Research Question

How can a credit-scoring pipeline combine multiple, independent drift signals to decide when to retrain, quantify whether retraining actually recovers performance under a leakage-free evaluation, and track whether the model's explanations stay trustworthy across that retraining cycle?

04 Related Work and Gaps Addressed

This project extends **"Fair and Explainable Credit-Scoring under Concept Drift"** (Shivogo, 2025, arXiv:2511.03807). The table below summarizes the approaches this work builds on and the specific limitation each one has that this project addresses.

Approach	Idea	Where it falls short
Static models (LogReg, XGBoost, LightGBM)	Train once, deploy	Performance quietly drops as data drifts
PSI / KS-test	Flag feature or label distribution shift	Primarily single-feature; misses interactions between features
Sliding-window retraining	Continuously retrain on recent data	Can overfit to short-term noise or transient patterns
Drift-triggered retraining	Retrain only when drift is detected	Effectiveness depends entirely on the detector's accuracy
SHAP explainability	Explain individual predictions	Feature importance itself can drift, complicating long-term interpretation

This project closes each of these gaps directly: it combines four independent drift-detection methods (not just PSI/KS) so that multivariate drift is caught even when no single feature crosses a threshold; it benchmarks the sliding window against two other retraining policies to confirm the windowing choice is justified rather than assumed; it couples the drift detector to the retraining trigger directly and reports when each individual detector would and would not have fired; and it tracks SHAP stability explicitly across every retrain, rather than computing it once.

05 Research Contributions

- **Multi-method drift detection.** PSI, a KS test on the label, Jensen-Shannon distance, and a classifier-based two-sample test are run together and compared, rather than relying on a single statistic family.
- **Leakage-free adaptive retraining evaluation.** The static and adaptive models are always scored on the identical held-out test split, with the adaptive model trained only on the complementary 80%, so the comparison cannot be inflated by evaluation mismatch.
- **Retraining-strategy benchmark.** Three retraining policies (most-recent-year, 3-year sliding window, full expanding window) are compared against the static baseline on the same test set, answering whether the sliding-window design is actually earning its complexity.

- **Threshold-aware evaluation.** AUC-ROC, PR-AUC (average precision), and F1 at both a naive 0.5 cutoff and a validation-tuned threshold are all reported together, so a misleading threshold artifact cannot be mistaken for a genuine performance difference.
- **SHAP stability monitoring.** Cosine similarity, Kendall-tau rank correlation, and top-10 Jaccard overlap track whether the model's explanations stay trustworthy as the data and the model itself change.
- **MLflow-tracked, reproducible pipeline plus an interactive dashboard** that presents every result and can run the full drift-detection and scoring pipeline live against a newly uploaded dataset.

06 Dataset

The Lending Club public loan dataset (2007–2018 release) is used as the source of real credit-risk data. After restricting to loans with a resolved outcome (Fully Paid, Charged Off, Default, or Late) and to the 2015–2018 issue-year range — the public release contains no loans issued 2019 onward — cleaning and feature engineering produce 919,696 records across 22 core columns, one-hot encoded to 84 model features using a schema fit once on the full cleaned dataset so every yearly window shares identical columns.

Year	Rows	Default rate
2015	377,184	20.53%
2016	298,553	24.69%
2017	178,817	27.21%
2018	65,142	27.17%
Total	919,696	–

The default rate itself rises steadily across the window — from 20.5% to 27.2% — which is a built-in label/prior shift useful for validating drift detection independently of any feature-level drift.

07 System Architecture and Methodology

The pipeline runs as five sequential phases, each producing artifacts consumed by the next.

Phase 1 — Preprocessing

Raw loan records are cleaned, the target is derived from loan status, numerical and categorical features are cast and imputed, and the data is split into one CSV per issue year (2015–2018). Categorical features are one-hot encoded using a single shared schema fit on the full cleaned dataset, guaranteeing that every yearly window has identical, correctly-aligned columns.

Phase 2 — Baseline Training

Logistic Regression, LightGBM, and XGBoost are trained on the 2015 window (SMOTE-balanced, MLflow-logged). XGBoost is selected as the reference model (AUC 0.7376 on a held-out 20% split). One additional XGBoost model is trained per subsequent year, purely for later per-year comparison.

Phase 3 — Drift Detection

PSI and Jensen-Shannon distance are computed per numerical feature against the 2015 reference; a KS test checks whether the label distribution (default rate) has shifted; and a classifier-based two-sample test trains a model to distinguish reference-year rows from current-year rows across all 84 features, using its cross-validated AUC as a multivariate drift signal. Retraining is triggered if any one of these methods fires.

Phase 4 — Adaptive Retraining and SHAP Analysis

For each drift year, the target year's data is split into an 80/20 train/test partition (stratified, fixed random seed). A validation slice is further carved from the training 80% to tune each model's decision threshold, without ever touching the test set. The adaptive model trains on a sliding window of recent years using only the training portion of the target year, and both the static and adaptive models are then scored on the identical held-out test set. SHAP values for the static model are computed on a sample of each year's data and compared to the 2015 baseline using cosine similarity, Kendall-tau, and top-10 Jaccard overlap.

Phase 5 — Backtesting and Reporting

The static and adaptive models are backtested across all years on matched held-out splits, producing the headline AUC-ROC-over-time figure, and a consolidated text report is assembled from every phase's saved output.

08 Drift Detection Results

All four drift-detection methods agree that drift is present from 2016 onward and grows in magnitude each year.

Year	Mean PSI	PSI≥critical	KS label drift	Mean JS dist.	Classifier AUC
2016	0.0129	0 features	Yes	0.0364	0.9769
2017	0.0352	1 feature (int_rate)	Yes	0.0634	0.9977
2018	0.0715	2 features (revol_util, int_rate)	Yes	0.0932	0.9989

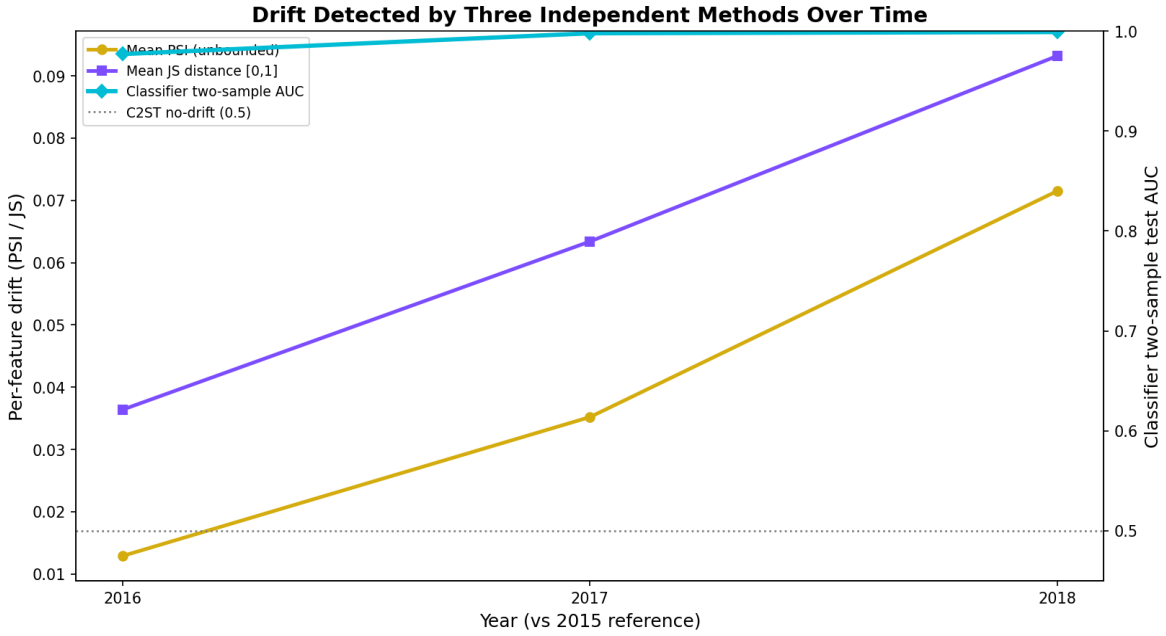


Figure 1. Mean PSI, mean JS distance, and classifier two-sample AUC over time, all measured against the 2015 reference.

The most important finding here is not visible from PSI alone: the classifier two-sample test reaches an AUC of 0.9769 by 2016, meaning a simple model can already tell 2016 data apart from 2015 data almost perfectly — yet PSI flags zero features as critically drifted that same year, and the single most-separable individual feature reaches only about 0.65 univariate AUC. The drift present in 2016 is therefore **multivariate**: no one feature has shifted enough to trip a per-feature threshold, but the joint distribution of features has shifted enough to be almost perfectly separable. Per-feature PSI alone would have badly understated this.

This has a direct operational implication. Under the deployed policy (retrain if *any* detector fires), drift triggers every year in this dataset, so drift-triggered retraining coincides with simple periodic yearly retraining here. A PSI-only policy, by contrast, would have skipped retraining in 2016 — missing a year where the classifier test and the label test both say drift is real. The trigger's practical value is this kind of avoidance of a false negative, not a reduction in retraining frequency on this particular dataset.

09 Cost of Model Staleness

A single XGBoost model trained once on 2015 data is evaluated, unmodified, on each subsequent year's full data to quantify how far performance falls if the model is simply left in production.

Year	N	Default rate	AUC-ROC	AUC drop	Drop %
2015	377,184	20.53%	0.7422	0.0000	0.00%
2016	298,553	24.69%	0.6948	0.0474	6.39%
2017	178,817	27.21%	0.6742	0.0680	9.16%
2018	65,142	27.17%	0.6695	0.0727	9.80%

By 2018, the static model has lost nearly a tenth of its discriminative power (0.7422 → 0.6695 AUC-ROC), directly quantifying the cost of never retraining.

10 Adaptive Retraining Results

For each drift year, the adaptive model (trained on a sliding window of recent years, using only the training portion of the target year) and the static 2015 model are scored on the identical held-out test split — the same rows, the same labels, nothing shared with training.

Year	Static AUC	Adaptive AUC	AUC gain	Static PR-AUC	Adaptive PR-AUC	PR-AUC gain
2016	0.6941	0.7123	+0.0182	0.4002	0.4405	+0.0403
2017	0.6769	0.7081	+0.0312	0.4106	0.4551	+0.0445
2018	0.6691	0.7222	+0.0531	0.4015	0.4756	+0.0741

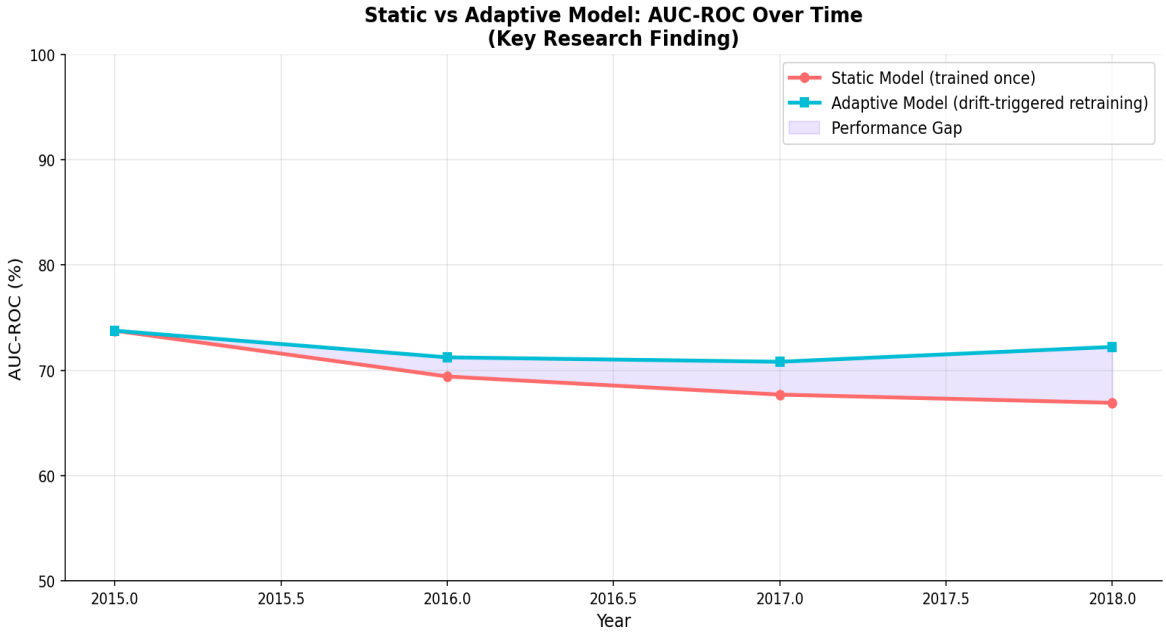


Figure 2. Static vs adaptive AUC-ROC over time. 2015 is the shared anchor before any retraining; the gap widens as drift accumulates.

The adaptive model outperforms the static baseline every year on both AUC-ROC and PR-AUC, and the size of the improvement grows as drift accumulates — the gain roughly triples from 2016 to 2018. PR-AUC (average precision) is reported alongside AUC-ROC because it is the more appropriate threshold-free metric for the imbalanced default class, and it shows the same pattern, with an even larger relative improvement.

A note on F1 and decision thresholds

At a naive fixed threshold of 0.5, the static model appears to have *higher* F1 than the adaptive model in most years — this is an artifact of the default rate rising over time (a fixed cutoff behaves differently as the positive-class base rate shifts), not evidence that the static model is better. When each model's decision threshold is instead tuned on a held-out validation slice (never the test set), the adaptive model's F1 overtakes the static model's every year: 0.484 vs 0.459 in 2016, 0.507 vs 0.481 in 2017, and 0.517 vs 0.476 in 2018. Both the naive and tuned F1 are reported side by side in the project's output files specifically so this artifact is visible rather than hidden.

11 Retraining Strategy Comparison

A natural objection to any adaptive-retraining result is that the gain might come from simply retraining at all, rather than from the specific sliding-window design. To test this, three retraining policies are benchmarked against each other and against the static baseline, all scored on the same held-out test set: retraining on only the most recent year (*naive-recent*), the deployed 3-year sliding window, and an *expanding* window that keeps all history from 2015 onward.

Year	Static AUC	Naive-recent AUC	Sliding AUC	Expanding AUC
2016	0.6941	0.7135	0.7123	0.7123
2017	0.6769	0.7128	0.7081	0.7081
2018	0.6691	0.7297	0.7222	0.7154

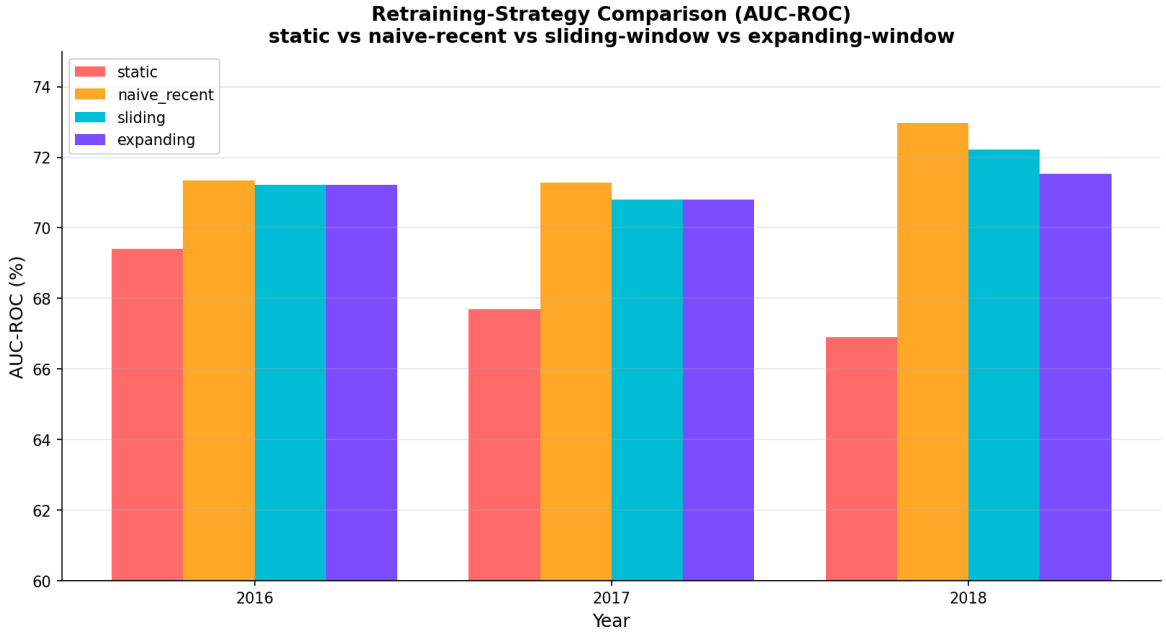


Figure 3. AUC-ROC by retraining strategy. By 2018 the strategies fan out clearly: naive-recent > sliding > expanding.

The central, and somewhat counter-intuitive, finding is that recency beats volume. The naive-recent strategy — using only the single most recent year of data — is the strongest performer every year, and by 2018 the expanding window (which keeps every year of history) has become the *weakest* of the three adaptive strategies (0.7154, against 0.7222 for the sliding window and 0.7297 for naive-recent). This shows that keeping stale 2015–2016 data in the training set actively hurts once drift has set in, rather than being harmless extra data. The deployed 3-year sliding window is best understood as a robust compromise: it captures most of the benefit of recency without requiring the optimal lookback window to be known in advance, which naive-recent implicitly assumes.

12 Explainability Stability (SHAP)

Beyond raw performance, a retrained model's *reasoning* matters for credit decisions that must be explained to applicants and regulators. SHAP values of the static 2015 model are computed on a

sample of each subsequent year's data and compared against the 2015 baseline using three complementary metrics.

Year	Cosine similarity	Kendall tau	Jaccard@10
2016	0.9931	0.9043	1.0000
2017	0.9806	0.8784	0.8182
2018	0.9784	0.8827	0.8182

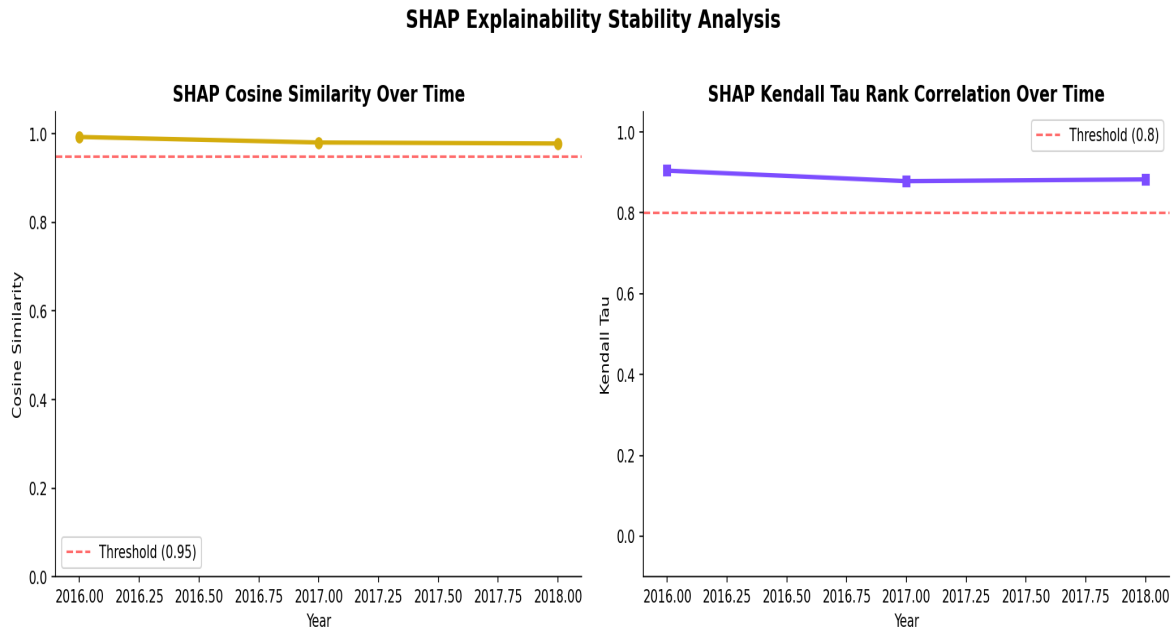


Figure 4. SHAP stability metrics over time (cosine similarity, Kendall tau, and top-10 Jaccard overlap vs the 2015 baseline).

Cosine similarity stays above the 0.95 stability threshold and Kendall tau stays above 0.8 every year, meaning the overall shape of the feature-importance vector is stable. Top-10 Jaccard overlap, however, drops to 0.818 in 2017 and 2018 — meaning the specific set of features in the top 10 has changed, even though the overall ranking correlation looks stable. The feature responsible is int_rate, which climbs into the top-10 by 2017–2018, directly tracking the same feature's PSI drift reported in Section 8. This is a useful early warning signal in its own right: an interpretability metric corroborating an independently-measured statistical drift signal.

13 Interactive Dashboard

All of the results above are also presented through an interactive Streamlit dashboard with six pages: an overview with headline KPIs, a drift-detection page with the multi-method comparison, a static-vs-adaptive comparison across AUC/PR-AUC/F1, a retraining-strategy comparison, an explainability page, and a live drift-check page. The live page lets a user pick a built-in yearly window or upload a new CSV, runs all four drift detectors against the 2015 reference in real time (roughly 10–40 seconds), and scores the data with both the static and adaptive models side by side. Built-in windows are scored on the same held-out test split used throughout this report, so the dashboard's numbers match this document exactly; uploaded data, having no leakage concern, is scored in full.

14 Limitations

- **No post-2018 data.** The Lending Club public release ends in 2018, so the COVID-era shock — likely the most extreme recent credit-market drift — cannot be tested; the drift observed here is the milder pre-2019 trend.
- **Single dataset, single market.** All results are on US unsecured consumer loans; the magnitude of degradation and recovery may not transfer to other portfolios or geographies.
- **Yearly granularity.** Real-world monitoring would typically run monthly; annual windows smooth over intra-year drift and limit the number of observable drift events.
- **Label latency.** Only loans with a resolved outcome (Fully Paid or Charged Off) are usable, under-representing the most recent originations — a survivorship/label-latency effect inherent to credit-risk data generally.
- **No probability calibration.** SMOTE balances the training data and thresholds are tuned on a held-out validation split, but the output probabilities themselves are not calibrated (e.g. via isotonic regression) and should not yet be read as true probabilities.
- **Trigger-frequency claim is untested.** Because drift fires every year on this dataset, the drift trigger cannot be shown here to reduce retraining frequency relative to a fixed periodic schedule — only to avoid retraining when one particular detector (PSI) is quiet. A longer, lower-drift series would be needed to demonstrate that benefit.
- **No confidence intervals.** All AUC/PR-AUC comparisons are reported as point estimates; a bootstrap over the test set would let the reported gains be stated with a confidence interval rather than as a single number.

15 Conclusion

This project demonstrates, on real lending data and under a leakage-free evaluation, that concept drift in credit-risk scoring is real, measurable by several independent methods, and costly if ignored (9.8% AUC lost by 2018). Drift-triggered adaptive retraining recovers most of that loss, with the recovery growing as drift accumulates. Benchmarking against simpler retraining strategies surfaced a genuine, non-obvious finding — that recency of training data matters more than its volume — rather than simply confirming the chosen design. Explainability was tracked throughout rather than computed once, showing that overall feature-importance stays statistically stable even while specific top-ranked features shift in ways that corroborate the independently measured drift. Together, the pipeline, the dashboard, and the honestly-reported limitations form a complete, reproducible answer to the research question posed in Section 3.

Reference: Shivogo, “Fair and Explainable Credit-Scoring under Concept Drift,” 2025, arXiv:2511.03807.